

Instructional Lab Guide for L1 Curriculum

All participants of L1 Course are requested to ensure that the following System configuration and Software are available.

It is fully based on tech stack (Java / Dotnet / Python) a candidate or Team is going to work with the use case.

Tech-Agnostic Full-Stack Development Using GenAI

1. Hardware / System Requirements

Minimum (Individual Developer Machine)

- **CPU:** Quad-core (Intel i5/AMD Ryzen 5 or above)
- **RAM:** 16 GB
- **Storage:** 512 GB SSD
- **GPU:** Not required for basic development, but useful if running local inference
- **OS Support:**
 - Windows 11
 - macOS 12+
 - Linux (Ubuntu 22.04 LTS recommended)

2. Core Software Requirements (Tech-Agnostic)

2.1. Back-End Development Requirements

(Choose any stack; requirements remain agnostic)

Languages (at least one)

- Node.js (LTS)
- Python 3.10+
- Java 17+
- .NET 8+
- Go 1.21+

2.2 Frameworks (typical options)

- Express / Fastify (Node)
- Django / FastAPI (Python)
- Spring Boot (Java)
- ASP.NET Core (C#)
- Gin / Echo (Go)

2.3 Package Managers

- npm / yarn / pnpm (Node)

- pip / conda (Python)
- Maven / Gradle (Java)
- NuGet (C#)

2.4 Database Clients

- PostgreSQL
 - MySQL
 - MongoDB
 - Redis
 - SQLite (lightweight dev)
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3. Front-End Development Requirements

3.1. Frameworks

- React
- Angular
- Vue
- Svelte

3.2 IDE

- Visual Studio Code
- Visual Studio

3.3 Extended Plugins

- Amazon Q Developer
- GitHub Copilot Chat

3.4. DevOps & Tooling - CI/CD

- GitHub Actions / Azure DevOps
- Jenkins (optional)

3.5. Security & Compliance Requirements

- API gateways
- OAuth / JWT

4. Testing Requirements

4.1 Unit Testing

- Jest / Mocha (JavaScript)
- PyTest (Python)
- JUnit (Java)
- xUnit (C#)